00846

Sri Lanka Institute of Information Technology

B.Sc. Special Honours Degree
in
Information Technology

Final Examination
Year 2, Semester 1 (2015)

Database Management Systems II (IT202)

Duration: 3 Hours

Instructions to Candidates:

- This paper has five questions. Answer **all** questions.
- Write answers in the booklet given.
- Total marks 100.
- This paper contains seven pages with the cover page.
- Calculators are not allowed.

08/04/2015

Question 1 **(20 marks)**

Consider a database schema consisting of the following relations for recording information on the clients of a stock broking firm:

Clients (<u>*clno*</u>, *name*, *phone*)
Stocks (<u>*scode*</u>, *sname*, *price*, *dividend*, *eps*)
BuySell (<u>*clno*, *trdate*</u>, *trtype*, *scode*, *qty*, *price*)
StocksOwned (<u>*clno*, *scode*</u>, *totalqty*, *totalcost*, *currval*)

A tuple in the Clients table consists of a unique client number, name and phone number. Each tuple of Stocks table contains unique stock code, stock name, the current price of a share, dividend per share, and earning per share (eps). Each transaction of a client which may be a buy or a sell of a stock is recorded a tuple in the BuySell table. It contains the client number, transaction date and time, type of transaction (B for buy or S for sell), stock code, number of shares (qty) and price per share for the transaction. The StocksOwned table contains a tuple for each stock a client currently owns, along with the total quantity of the stock, the total cost of purchasing the stock, and its current value. The primary key of each table is underlined.

a) Express the following queries in SQL.
   i. Find the name and price of each stock that is owned by five or more clients.
   (4 marks)

   ii. Find the name and phone number of clients who have not bought any stock since 1st January 2014.
   (5 marks)

b) Write T-SQL procedure named *insertasell* to insert a row into the BuySell table for a sell transaction given the input parameters of client number, stock code, and quantity. The transaction date and time should be the current date and time; the price is the current price of the stock in the Stocks table.
(5 marks)

c) Assume that a procedure named *updstockowned* is already available to update the *StocksOwned* table. It takes clients number, stock code, quantity, price, and transaction type as a parameters. When the transaction type is 'B', the procedure increase the total quantity and cost of the given stock for the client (or inserts a new row for client and stock if that client did not own the previously).

Write a trigger named *buytrigger* on the BuySell table that will update the StocksOwned table by calling this procedure, whenever a new row of a Buy transaction is inserted into the BuySell table. [Only write the trigger, don't write the updstockowned procedure]
(6 marks)

Question 2 **(20 marks)**

a) The time spent in accessing a data block in a disk,

Access time = Seek time + Rotational delay + Data transfer time

Briefly explain two solutions to minimize the access time.

(5 marks)

b) Consider a disk with a sector size of 512 bytes, a block size of 1024 bytes, 2000 tracks per surface, 50 sectors per track, 5 double-sided platters, average seek time of 10msec.

i. What is the capacity of a track in blocks?

(1 mark)

ii. What is the capacity of a cylinder in blocks?

(1 mark)

iii. What is the capacity of the disk in blocks?

(1 mark)

iv. If disk platter rotates at 5400 rpm (revolutions per minute), what is the average rotational delay?

(2 marks)

v. What is the access time to read two consecutive disk blocks?

(2 marks)

c) Describe the *sequential flooding* of the buffer pool?

(3 marks)

d) Consider the following buffer pool with 5 frames.

Frame No: 0	Frame No: 1	Frame No: 2	Frame No: 3	Frame No: 4
PageID: 5	PageID: 20	PageID: 30	PageID: 4	PageID: 15
PinCount: 1	PinCount: 1	PinCount: 1	PinCount: 0	PinCount: 0
Dirty: 1	Dirty: 1	Dirty: 0	Dirty: 0	Dirty: 0

Each frame has a frame number (*Frame No*), page id (*PageID*), pin count (*PinCount*) and a dirty bit (*Dirty*).

Assume that the following times states the last time the particular frame was accessed (on the same day):

Frame number 0 at 11:15 am
Frame number 1 at 11:35 am
Frame number 2 at 10:57 am
Frame number 3 at 11:12 am
Frame number 4 at 11:26 am

Describe the steps involved when a page request for the page with PageID 32 occurs if MRU (most-recently-used) policy is used as the replacement policy.

(5 marks)

Question 3 **(18 marks)**

a) Compare heap file against the sorted file organization.

(3 marks)

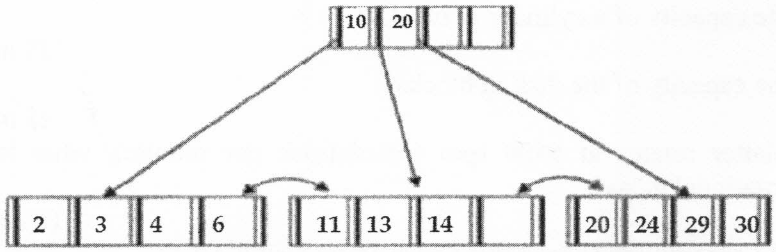
b) Consider an **employee** relation containing records with employee id, employee name, age and salary. Assume that the records are stored in a hash file where the hash function is based on the employee's **salary**.

"*Retrieving records with **age** of 20 is inefficient with the above file*"

State whether the above statement is true or false. Give reasons.

(3 marks)

c) Consider the B+ tree index given below.



i. Illustrate the tree after inserting 31.

(2 marks)

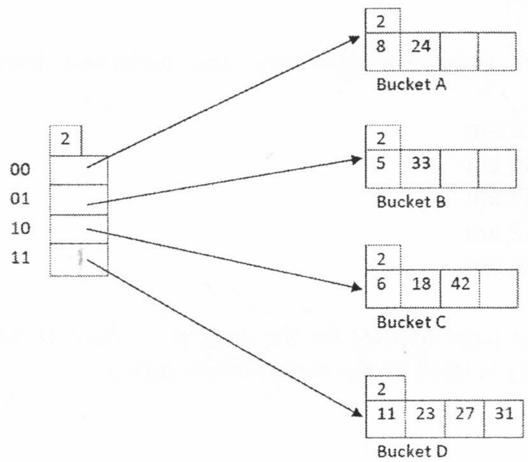
ii. Illustrate the tree after deleting 20, 24, 29 from the original tree.

(2 marks)

iii. Construct an order 2 B+ tree index when the following index entries are inserted in the given order.
3, 5, 15, 12, 7, 2, 20, 22, 25

(4 marks)

d) Consider the Hash index below;



Draw the hash index after inserting 9, 17 and 37.

(4 marks)

Question 4 **(18 marks)**

a) Consider the following statement;
"Indexes speed up selections but may slow down updates."
Briefly explain how indexes can "slow down" update operations.

(2 marks)

b) Justify why a table contains only one clustered index whereas more unclustered indexes are allowed.

(3 marks)

c) Consider the following schema:
Employee(<u>*eid*</u>:integer, *ename*:string, *address*:string, *salary*:float, *dno*:int)

There are 10000 pages in Employee file. Each page contains 100 tuples.
Assume that the following indexes exists:
Index 1: An unclustered B+ Tree index on Employee<eid> with height 3.
Index 2: A clustered hash index using Alt. 1 on Employee<eid> (Assume a cost of 1.2 Disk I/Os to access a hash page).

Consider the following queries:

Query 1:	SELECT	salary
	FROM	Employee
	WHERE	eid = 'E100200762'
Query 2:	SELECT	salary
	FROM	Employee
	WHERE	eid < 'E00001000'

i. What is the best index (from the indexes given above) that you would select to evaluate **query 1**? Give reasons? Estimate the cost of the query in disk I/Os using the selected index. Note that *eid* is primary key in the Employee table.

(4 marks)

ii. What is the best index (from the indexes given above) that you would select to evaluate **query 2**? Give reasons?

(2 marks)

d) Consider the following relational schema:
Student (<u>sno</u>, sname, gpa)
Registration (<u>sno, cno</u>, marks)
Course (<u>cno</u>, cname, credits)

Assume that Student relation consists of 300 pages with 100 tuples per page, Registration relation contains 900 pages with 100 tuples per page, and Course relation contains 300 pages with 100 tuples per page.

Consider the following query:

Select	count (s.sno)
From	Student s, Registration r
Where	s.sno = r.sno

There exists a B+ tree on Registration<sno> with height 3 using Alt. 1 (record and index key together). Join tables using index nested loop join algorithm.

i. Draw query tree for the above query.
(2 marks)

ii. Estimate the cost for the above query.
(5 marks)

Question 5 (24 marks)

a) Briefly explain the properties of a transaction.
(4 marks)

b) Compare *serial* and *serializable* schedules.
(3 marks)

c) Give an example for a schedule that is unrecoverable. Briefly explain the schedule.
(3 marks)

d) Briefly explain the *deadlock detection* algorithm.
(3 marks)

e) Consider the following part of the schedule.

T_1	T_2	T_3	T_4
S(A)			
R(A)			
		S(C)	
		R(C)	
S(C)			
	X(B)		
	W(B)		
		X(B)	
			X(D)
			W(D)
	X(D)		
			X(A)

Assume that Transaction T_i is higher priority than transaction T_{i+1}(i.e. transaction T_1 has higher priority than T_2; T_2 has higher priority than T_3; and T_3 has higher priority than T_4).

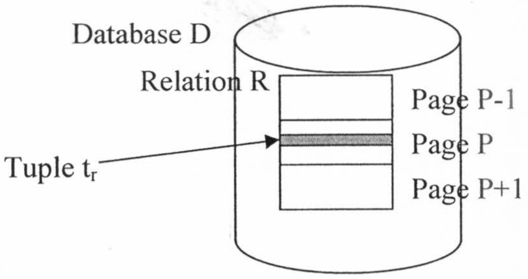
i. Draw a **Wait-For-Graph** for the above given schedule.
(3 marks)

ii. What should be done next for deadlock detection approach to break the deadlock after identifying it?
(2 marks)

f) Briefly explain Simple Tree Locking algorithm.

(3 marks)

g) Consider the following scenario:



Database D contains a relation R. Relation R contains a page P. Page P contains a tuple t_r.
Assuming that multiple granularity locking scheme is used, describe the locks acquired when writes all tuples of page P.

(3 marks)

-- End of the Question Paper --

08/04/2015